How to install and remove software using the APT command. Include several well documented examples as you will need to install software later.

- **What APT is:** It's the tool Ubuntu, Debian, and Mint use to install software. Most commands need `sudo` because they change the system.
- **Always update first:** `sudo apt update` refreshes the list of available software. It doesn't install anything by itself.
- **Installing a program:** `sudo apt install <package-name>`
  - Example: `sudo apt install tree`
  - Type `y` when it asks if you want to continue
- **Skipping the prompt:** Add `-y`, like `sudo apt install -y htop`
- **Installing a few things at once:** `sudo apt install -y git curl nano`
- **Finding software:** `apt search <keyword>` looks for packages, and `apt show <package-name>` gives details about one
- **Checking that it worked:**
  - `which tree` shows `/usr/bin/tree` if it's installed
  - `tree --version` shows the version number
- **Removing software:**
  - `sudo apt remove tree` removes the program but leaves its settings behind
  - `sudo apt purge tree` removes the program and its settings
  - `sudo apt autoremove` cleans up leftover files nothing else needs
- **Upgrading everything:** `sudo apt update && sudo apt upgrade -y`
- **If something goes wrong:**
  - "Unable to locate package" means run `sudo apt update` and check your spelling
  - "Could not get lock" means another update is running, so wait a minute and try again
  - "Permission denied" means you forgot `sudo`
- **The one thing to remember:** Run `sudo apt update` before you install anything

How to create a shell script step by step including screenshots and how to run it. Try to be as detailed as possible. You will have to create a script in your final exam.

- **Open the terminal (Debian 13)**
  - Click **Activities**, type "Terminal," and press Enter
  - Or open the **Files** app, right-click an empty spot, and choose **Open in Terminal**
  - *Screenshot: the open terminal window*
- **Open the manual page for free**

- `man free`
- *Screenshot: the man page for free*
- If you get "man: command not found," install it: `sudo apt install man-db`

- **Moving around inside a man page**
  - `Space` goes down one page, `b` goes back one page
  - `Up` and `Down` arrows scroll line by line
  - `/word` searches for a word (example: `/human`), then `n` jumps to the next match
  - `q` quits and returns to the terminal

- **What free does:** shows how much memory (RAM) and swap is used and available
  - `free` shows values in kibibytes
  - `free -h` shows human-readable sizes (MB, GB)
  - `free -m` shows megabytes
  - `free -g` shows gigabytes
  - `free -t` adds a totals line
  - `free -s 2` refreshes every 2 seconds (stop with `Ctrl + C`)
  - *Screenshot: output of free -h*

- **Reading the free output**
  - **total:** all the memory installed
  - **used:** memory in use
  - **free:** memory not being used at all
  - **available:** memory programs can still use (the most useful number)
  - **Swap:** disk space used as backup memory

- **Commands for finding what you need**
  - `man <command>` opens the full manual (example: `man ls`)
  - `man -k <keyword>` or `apropos <keyword>` searches all man pages by keyword (example: `man -k memory`)
  - `whatis <command>` gives a one-line description (example: `whatis free`)
  - `<command> --help` shows a quick summary of options (example: `free --help`)
  - `info <command>` opens the GNU info manual for some commands
  - `which <command>` shows where a program is located
  - `whereis <command>` shows the program and its man page location
  - `type <command>` tells you if it's a program, alias, or built-in

- **Commands for finding files**
  - `find ~ -name "hello.sh"` searches your home folder for a file by name
  - `ls` lists files, `ls -l` shows details, `ls -a` shows hidden files
  - `pwd` shows the folder you're in
  - `cd <folder>` moves into a folder, `cd ..` goes up one level, `cd ~` goes home
  - `grep "text" <file>` searches inside a file for text

- **Commands for checking your system**

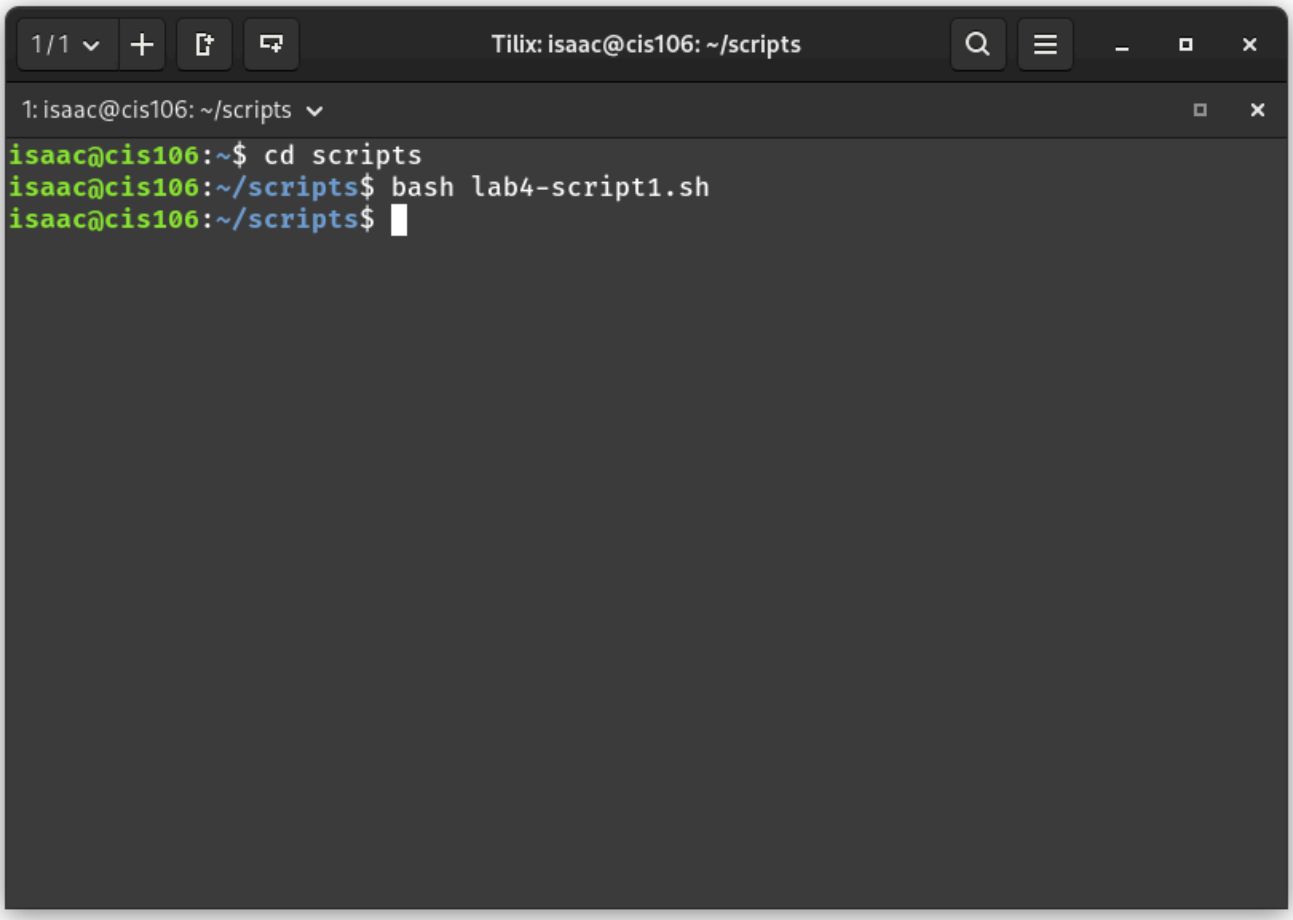
    - `df -h` shows disk space
    - `du -sh <folder>` shows how big a folder is
    - `uptime` shows how long the system has been running
    - `top` shows running processes (press `q` to quit)
    - `lscpu` shows CPU information
    - `uname -a` shows kernel and system information
    - `whoami` shows your username
    - `date` shows the date and time

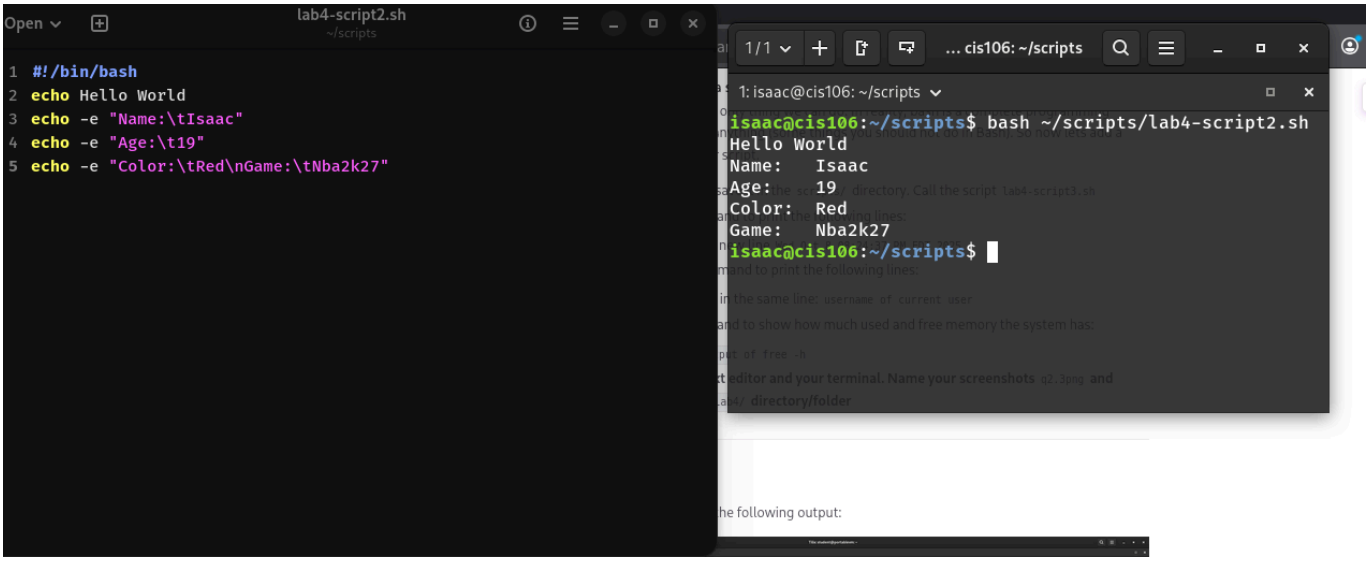
- **Handy terminal tricks**

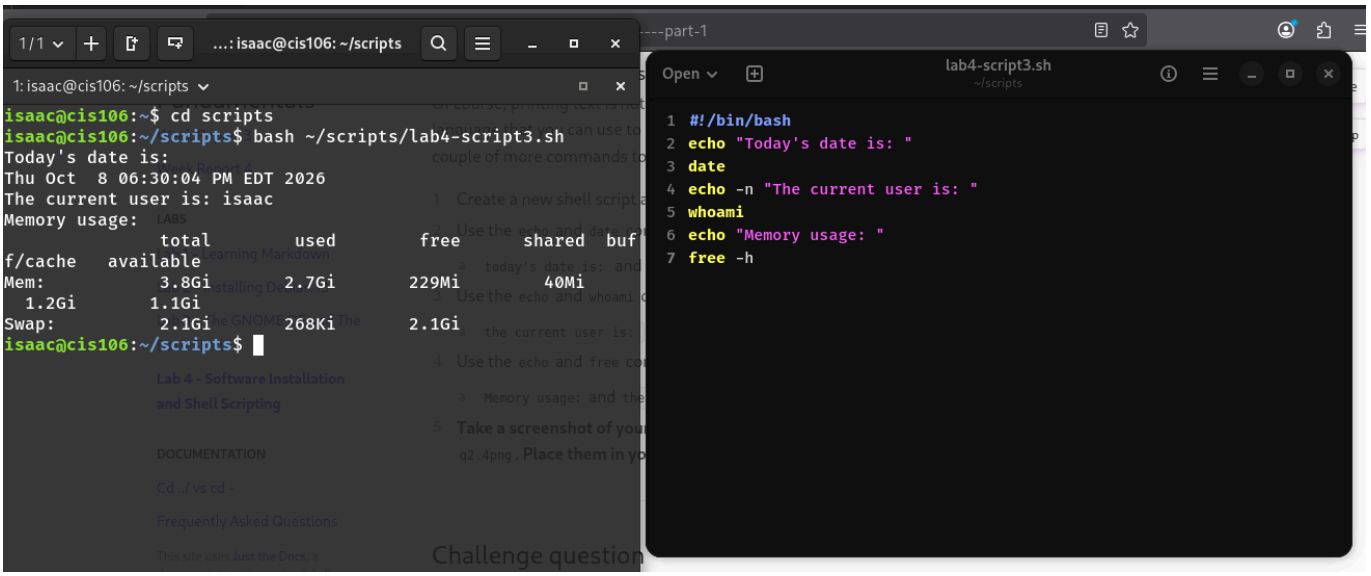
    - `Tab` auto-completes file and command names
    - `Up arrow` brings back the previous command
    - `Ctrl + C` stops a running command
    - `clear` clears the screen
    - `history` lists the commands you've typed

Screenshots

Question 2


1 / 1
Tilix: isaac@cis106: ~/scripts
1: isaac@cis106: ~/scripts
isaac@cis106:~$ cd scripts
isaac@cis106:~/scripts$ bash lab4-script1.sh
isaac@cis106:~/scripts$


Open
lab4-script2.sh
~/scripts
1 #!/bin/bash
2 echo Hello World
3 echo -e "Name:\tIsaac"
4 echo -e "Age:\t19"
5 echo -e "Color:\tRed\nGame:\tNba2k27"
1 / 1
... cis106: ~/scripts
1: isaac@cis106: ~/scripts
isaac@cis106:~/scripts$ bash ~/scripts/lab4-script2.sh
Hello World
Name: Isaac
Age: 19
Color: Red
Game: Nba2k27
isaac@cis106:~/scripts$


1/1
...: isaac@cis106: ~/scripts
1: isaac@cis106: ~/scripts
isaac@cis106:~$ cd scripts
isaac@cis106:~/scripts$ bash ~/scripts/lab4-script3.sh
Today's date is:
Thu Oct 8 06:30:04 PM EDT 2026
The current user is: isaac
Memory usage:
total used free shared buf
f/cache available
Mem: 3.8Gi 2.7Gi 229Mi 40Mi
1.2Gi 1.1Gi
Swap: 2.1Gi 268Ki 2.1Gi
isaac@cis106:~/scripts$
Open
lab4-script3.sh
~/scripts
1 #!/bin/bash
2 echo "Today's date is: "
3 date
4 echo -n "The current user is: "
5 whoami
6 echo "Memory usage: "
7 free -h

Question 3

Open challenge_lab4.sh ~/cis106/labs/lab4

```
#!/bin/bash
printf '————————————————————————————————————————\n'
figlet "system"
echo
echo -e "DATE UTC:   $(date -u '+%m/%d/%y %H:%M:%S')"
echo -e "HOSTNAME:   $(hostname)"
echo -e "UPTIME:     $(uptime -p)"
echo -e "USER:   \t$(whoami)"
figlet "memory"
echo
echo -e "Mem:    \t$(free -m -h)"
figlet "DISK"
df -h /
printf '————————————————————————————————————————\n'
```

1/1 Tilix: isaac@cis106: ~/cis106/labs/lab4

1: isaac@cis106: ~/cis106/labs/lab4

```
isaac@cis106:~/cis106/labs/lab4$ bash challenge_lab4.sh
————————————————————————————————————————
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\__ \ |_| \__ \ ||  __/ | | | | |
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        |___/

DATE UTC:       10/09/26 00:26:29
HOSTNAME:       cis106
UPTIME:         up 2 hours, 54 minutes
USER:           isaac

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|_| |_| |_|\___|_| |_| |_|\___/|_|   \__, |\__,_| |
                                     |___/

Mem:                           total        used        free      shared  buff/cache   available
Mem:           3.8Gi       2.4Gi       575Mi        40Mi       1.2Gi       1.5Gi
Swap:          2.1Gi       445Mi       1.6Gi
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|  _ \_ _/ ___| |/ /
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| |_| | | ___) | . \
|____/___|____/|_|\_\

Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        38G   11G   25G  30% /
————————————————————————————————————————
isaac@cis106:~/cis106/labs/lab4$
```